A choice-free Lean formalization of Bishop–Cheng dominated convergence over presented reals

Author

Toshihisa Urahigashi
Independent researcher
Nara, Japan
123026.urahigashi.toshihisa@gmail.com
ORCID 0009-0004-0460-6242

Corresponding author

Toshihisa Urahigashi, 123026.urahigashi.toshihisa@gmail.com

Author contributions

The paper has a single author, who conceived the work, carried out the formalization, built and ran the audit apparatus, and wrote the manuscript, and who is responsible for all of it.

Funding

No funding was received for conducting this study.

Competing interests

The author has no competing interests to declare that are relevant to the content of this article.

Data and code availability

There is no dataset. The Lean 4 artifact, the audit apparatus and the shipped audit logs are archived and citable at `10.5281/zenodo.22330020`, and are also at `https://github.com/turahigashi/choicefree-bc-dct` under tag `v0.7.7`, an annotated tag pointing at the commit the deposit was cut from. Reproducing the build requires Lean 4.30.0 and mathlib 4.30.0; the remaining checks need only Python 3 and `coreutils`.

Ethics approval

Not applicable.

Preprint

Two earlier drafts of this manuscript, under the title "A choice-free Lean formalization of a profile-based dominated convergence theorem in presented Bishop–Cheng measure theory", were deposited on Zenodo as preprints under CC BY 4.0 on 2026-08-09 (`10.5281/zenodo.21859192`) and 2026-08-13 (`10.5281/zenodo.21911760`); the concept DOI is `10.5281/zenodo.21859191`. Neither has been peer reviewed, and the present manuscript supersedes them. The work has not been posted to arXiv. The Lean 4 artifact it reports, together with the audit apparatus and the shipped audit logs, is archived separately under the MIT licence at `10.5281/zenodo.22330020`.